

The Adelic Distinction: Physics as Automorphic Representation Theory on the Idele Class Group

Rowan Brad Quni-Gudzinis

2026-08-04

Abstract

Ostrowski's theorem (1916) establishes that every completion of the rational numbers $\mathbb{Q}$ is either the real Archimedean place $\mathbb{R}$ or a p -adic non-Archimedean place $\mathbb{Q}_p$. Physics has, for a century, restricted itself to the Archimedean completion alone — using real manifolds, continuous groups, and dimensional constants — while the p -adic places have remained an unutilized mathematical structure. This paper proposes that the adèle ring $\mathbb{A}_{\mathbb{Q}}$ and its unit group, the idele class group $\mathbb{C}_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$, provide the single mathematical object whose automorphic representations generate all known particle physics, gauge interactions, and cosmological parameters. Within this framework: (i) the circle S^1 underlying $U(1)$ gauge symmetry is the Archimedean projection of the unit idele class group; closure and quantization follow from the product formula $\prod_v |q|_v = 1$; (ii) spin is a character of adelic rotations on Bruhat-Tits trees; (iii) the three generations of fermions arise from the tripartite structure of the global idele class group; (iv) quantum field theory becomes automorphic harmonic analysis on the adèles, with UV divergences cancelled by telescoping sums across all places; (v) the cosmological constant's smallness emerges as the Euler characteristic of the adelic symmetric space rather than a 10^{-120} fine-tuning; (vi) the Riemann hypothesis is equivalent to the unitarity of adelic distinction dynamics. Experimental signatures — prime-number gravitational wave echoes, parity-violating CMB spectra, and collider fractal substructure — are proposed as falsifiable predictions. The framework resolves the mutual isolation of number theory, quantum foundations, and cosmology that has persisted for 66–110 years across five independent disciplines.

Keywords: adèles, idele class group, Bruhat-Tits tree, Ostrowski's theorem, automorphic representations, distinction network, product formula, p -adic physics, cosmological constant, Riemann hypothesis

1. Introduction

1.1 The Archimedean Assumption

The standard formalism of physics rests on an unstated assumption: that the real numbers $\mathbb{R}$ are the correct number system for describing physical quantities. Coordinates, field values, coupling constants, and probability amplitudes are all taken to be real numbers. The real numbers enter physics through the back door of continuity — the assumption that spacetime is a smooth manifold, that symmetry groups are Lie groups, and that functional integrals run over real-valued fields.

Ostrowski's theorem [1] shows that this choice is not forced by logic. The rational numbers $\mathbb{Q}$ admit completions at every prime p , yielding the p -adic fields $\mathbb{Q}_p$, in addition to the real completion $\mathbb{R}$. The theorem is exhaustive: these are all the completions. Physics has, without justification, restricted itself to the real place alone. This paper asks what happens when the restriction is lifted — when physical quantities are permitted to carry p -adic components, and the natural arena is the adèle ring $\mathbb{A}_{\mathbb{Q}}$, the simultaneous product of all completions.

The question is not speculative in the sense of proposing new physical laws; it is structural. The adèle ring already exists as a mathematical object, independent of any physical theory. The claim is that its structure alone — the product formula, the topology of the idele class group, the geometry of Bruhat-Tits trees — constrains the form that any physical theory defined over it must take, and that those constraints reproduce the central features of the Standard Model and cosmology without additional assumptions. [speculative]

1.2 The Distinction Network

The framework adopted here is that the physical world is a dynamical network of distinctions [2]. A distinction is a primitive act of differentiation — the drawing of a boundary that separates one state from another. The set of all possible distinctions over the rational numbers $\mathbb{Q}$ naturally organizes into the adèle ring: the Archimedean place encodes continuous distinctions (real frequencies), while each p -adic place encodes discrete distinctions indexed by prime numbers. The totality is the adelic distinction network, and its invariant structure — the idele class group — is the source of all conserved quantities and symmetries. [my conjecture]

This paper does not attempt to derive dynamics from an action principle or to quantize a classical theory. Instead, it enumerates the structural consequences of the adelic topology and shows that they align with observed physics across an unexpected range of phenomena. The approach is exploratory: it maps the landscape, identifies correspondences, and proposes falsifiable tests. The derivation of numerical constants — masses, couplings, mixing angles — as periods of automorphic forms is left as a program for future work.

2. The Adelic Circle

2.1 The Unit Idele Class Group

The familiar circle $S^1 = \{e^{i\theta} : \theta \in [0, 2\pi)\}$ is the group of complex numbers of absolute value 1 at the real place. But each p -adic place also possesses a “unit circle”: the group $\mathbb{Z}_p^\times$ of p -adic integers with p -adic absolute value 1. The full adelic circle is the restricted product of all these local unit circles: the unit idele class group $\mathbb{C}_{\mathbb{Q}}^{(1)}$, a compact abelian group that is a hybrid of the continuous real circle and the totally disconnected profinite product of p -adic unit groups.

The product formula,

$$\prod_v |q|_v = 1 \quad \text{for all } q \in \mathbb{Q}^\times,$$

is the statement that a rational number’s “size” across all completions multiplies to unity. In physical terms, this is the closure condition for adelic cycles: a loop in the distinction network returns to its starting point because the product of all local valuations is conserved. The Archimedean winding number (continuous phase) is balanced by p -adic torsion (discrete phases), and the total invariant is an element of $\mathbb{Z}$, the fundamental group of the adelic circle. [speculative]

2.2 Gauge Theory as Archimedean Projection

The $U(1)$ gauge group of electromagnetism is precisely the Archimedean connected component of the unit idele class group. The full gauge group, were the p -adic places included, would be the entire idele class group — an extension of $U(1)$ by a profinite group encoding discrete symmetries. The restriction to $U(1)$ alone is a provincial simplification: it discards the p -adic components that may govern parity, charge conjugation, time reversal, and the generation structure of fermions. [speculative]

A Wilson loop in gauge theory, measuring the holonomy around a closed spacetime path, is an element of $U(1)$. In the adelic framework, it is an idele class group element evaluated along an adelic cycle. Its value is a pure phase — an element of the unit idele class group — because the product formula enforces global invariance. Gauge invariance is not an additional postulate; it is the statement that physical observables are invariant under the product formula. [my conjecture]

3. Particles as Automorphic Representations

3.1 Spin as Adelic Rotation

In quantum mechanics, a fermion’s wavefunction acquires a phase of -1 under a 2π spatial rotation. In the adelic picture, a rotation is not a smooth continuous motion but a cyclic permutation of distinction edges at a vertex of the Bruhat-Tits tree. At a prime p , the tree is a $(p+1)$ -regular graph. A rotation corresponds to a permutation of the $p+1$ incident branches. The global phase acquired under a full rotation is a

character of the idele class group evaluated on this cycle, yielding -1 for fermions and $+1$ for bosons — determined by the parity of edge permutations across all primes, enforced by the product formula. [speculative]

The spin-statistics connection thus becomes a corollary of the adelic rotation character: exchange of two identical particles corresponds to a half-rotation in adelic space, producing the same phase as a full rotation, so fermions antisymmetrize and bosons symmetrize.

3.2 Charge Quantization

Why is electric charge quantized in integer multiples of e ? In the adelic framework, charge is the winding number of the particle's Compton idele around the unit idele class group $\mathbb{C}_Q^{(1)}$. The character group (Pontryagin dual) of this compact abelian group is $\hat{\mathbb{Z}}$, the profinite completion of the integers, plus a real winding number. A particle's electromagnetic coupling is a continuous homomorphism from $\mathbb{C}_Q^{(1)}$ to $U(1)_{\text{real}}$, classified by an integer — the electric charge. There is no continuous deformation; the charge is topologically locked by the compactness of the unit idele class group. [speculative]

3.3 Three Generations

The existence of three fermion generations — (e, μ, τ) , (u, c, t) , (d, s, b) — with identical quantum numbers but hierarchical masses, has no compelling explanation in the Standard Model. The adelic framework suggests that the three generations correspond to the three non-trivial real characters of the idele class group's connected component, or more precisely, to a triple product structure in the global automorphic spectrum [3]. The cubic extension of $\mathbb{Q}$ associated with the absolute Galois group's action on a three-dimensional vector space over a finite field yields exactly three independent automorphic forms matching the chiral fermion structure. The mass hierarchy $m_e \ll m_\mu \ll m_\tau$ emerges as the exponentiation of the prime spectrum's decay: the Yukawa couplings are periods of a motive with Hodge numbers $(1, 1, 1)$, evaluated at different primes. [speculative]

4. Quantum Field Theory as Automorphic Harmonic Analysis

4.1 Fields as Automorphic Forms

If spacetime is the emergent shadow of the adelic trees, then the primary entities are not operator-valued distributions on a manifold but automorphic forms on the idele class group. Every particle species corresponds to an irreducible automorphic representation of a reductive group over the adeles. Creation and annihilation operators are Fourier coefficients of these automorphic forms, with the Fourier expansion over the Pontryagin dual of the adelic circle.

The path integral over fields in spacetime becomes a sum over automorphic representations, weighted by their L -functions. Feynman diagrams are combinatorial expansions of the adelic action's partition function, with the propagator given by the inverse of the Laplacian on the adelic symmetric space. [speculative]

4.2 Finiteness from the Adelic Topology

The renormalization group flow — the modulation of couplings with energy scale — is the scale evolution along the Bruhat-Tits tree from the Archimedean boundary into the bulk. The beta function is the derivative of the idele's valuation with respect to the scale parameter. Crucially, UV divergences are absent: integrals over the real continuum are replaced by adelic integrals where the non-Archimedean components are compact and finite-volume. The vacuum energy E_{vac} is an adelic integral:

$$E_{\text{vac}} = \sum_v \int \|k\|_v d\mu_v(k)$$

where $d\mu_v$ is the Haar measure on the local field at place v . The Archimedean place gives the usual quartic divergence; the p -adic integrals yield negative contributions due to the ultrametric topology. The product formula forces a telescoping cancellation across all places, leaving a residual mismatch equal to the Euler characteristic of the adelic symmetric space. This mismatch is the observed dark energy density — not 10^{-120} of the Planck density by fine-tuning, but naturally small because the adelic cancellation is exact at leading order. [speculative]

5. A Unified View of Core Symmetries

5.1 CPT as the Product Formula

CPT symmetry — the combination of charge conjugation, parity inversion, and time reversal — is a core invariance of any Lorentz-invariant quantum field theory. In the adelic framework, CPT is the reflection symmetry of the full adelic distinction network: parity inverts the orientation of each Bruhat-Tits tree, charge conjugation conjugates winding numbers, and time reversal inverts the direction of scale flow. The product of these three involutions is an automorphism of the idele class group that leaves the action invariant. The functional equation of the adelic L -function is precisely the mathematical statement of CPT invariance: the theorem that CPT must be conserved follows from the product formula. [speculative]

5.2 The Higgs as Dilaton of the Adelic Symmetric Space

The Higgs field gives mass to particles via spontaneous symmetry breaking. In the adelic picture, the distinction network possesses a global scale symmetry — the product formula ensures balance across all valuations. The Higgs field is the dilaton of the adelic symmetric space: the field that parametrizes the relative scale between the Archimedean and p -adic completions. Its potential is generated by an adelic anomaly: the product formula is exact for the global object, but local fluctuations around the vacuum can break it. The Higgs vev sets the absolute scale of particle

masses by fixing the conversion factor between the real norm and the p -adic valuations. The quadratic divergence of the Higgs mass in standard QFT is cancelled by the telescoping sum across all places. [speculative]

6. Dark Matter, Neutrinos, and the Strong CP Problem

6.1 Dark Matter as Torsion in the Idele Class Group

Dark matter consists of automorphic forms whose p -adic components carry non-trivial torsion under the unit idele class group. Their interactions with ordinary matter are suppressed because the Archimedean coupling is small (they are “near” the trivial character), but they gravitate because the stress-energy of the full adelic representation is non-zero. The stable dark matter particle corresponds to the torsion subgroup’s generator — its Compton number is a pure root of unity in the p -adic place and a very small real number, explaining why dark matter is cold and collisionless. [speculative]

6.2 Neutrino Masses

The right-handed neutrino is a purely non-Archimedean automorphic form — its Compton idele has zero Archimedean component, living entirely on the Bruhat-Tits trees. Its mass is set by the scale of the deepest p -adic branch, near the Planck scale. The seesaw mechanism $m_\nu \sim v^2/M_R$ follows directly, with v the Higgs vev and M_R the p -adic mass scale. The PMNS mixing matrix arises from misalignment between the adelic basis of charged leptons and the automorphic basis of neutrinos — a set of adelic periods. [speculative]

6.3 The Strong CP Problem

QCD permits a CP-violating term $\theta G\tilde{G}$ with θ experimentally constrained to $< 10^{-10}$. The Peccei-Quinn mechanism introduces an axion field that dynamically relaxes θ to zero. In the adelic framework, the axion is the modulus of the unit idele class group — a field parametrizing the relative phase between the Archimedean circle and the product of p -adic unit circles. The θ angle is an adelic phase, and the product formula forces the total phase to be an integer multiple of 2π , meaning the observable θ (the Archimedean projection) is zero in the vacuum. The axion is the pseudo-Goldstone boson of the product formula’s accidental symmetry. [speculative]

7. Quantum Gravity and Cosmology

7.1 The Riemann Hypothesis as Unitarity

The zeros of the Riemann zeta function have long been conjectured to be eigenvalues of a quantum chaotic Hamiltonian (Hilbert-Pólya conjecture). In the adelic framework, the distinction graph's Laplacian on the adelic symmetric space is a self-adjoint operator whose spectrum decomposes into automorphic representations. The Selberg trace formula relates the spectrum to lengths of closed geodesics on the adelic space — precisely the primes. The zeta function is the spectral determinant of this Laplacian.

Consequently, the Riemann hypothesis — that all non-trivial zeros lie on the critical line $\Re(s) = 1/2$ — is equivalent to the statement that the distinction network's Hamiltonian is Hermitian (having real eigenvalues). If the distinction dynamics is unitary, the zeros must lie on the critical line. The physical universe's unitarity and the Riemann hypothesis are the same statement. [speculative]

The spacing distribution of zeta zeros follows the Gaussian Unitary Ensemble (GUE) of random matrix theory, characteristic of quantum chaotic systems without time-reversal symmetry. In the distinction network, time-reversal symmetry is broken by the p -adic orientation (the tree is directed by scale flow), producing the observed GUE statistics. [speculative]

7.2 Time as Renormalization Group Flow

Time's arrow comes from the irreversible accumulation of distinctions. In precise terms: time is the adelic renormalization group flow. The scale parameter on each Bruhat-Tits tree is the number of branchings from the boundary. Moving inward (toward the root) coarse-grains the distinction network — an irreversible process generating entropy. The real time we experience is the flow parameter of the Archimedean place, Wick-rotated to Lorentzian signature. The Wheeler-DeWitt equation — the timeless Schrödinger equation of quantum gravity — is replaced by the adelic Dyson-Schwinger equation, a constraint on the full automorphic state from which time emerges as gradient flow toward increasing distinction entropy. [speculative]

The initial singularity is the point where all trees are at infinite depth (scale zero) — the state before any distinctions are drawn. The first distinction immediately generates the entire adelic tree structure in a cascade. The arrow of time is the direction of increasing tree depth. [speculative]

7.3 Dimensionality from Spectral Geometry

Why does spacetime have 3+1 large dimensions? The adelic symmetric space for $\mathrm{SL}(2, \mathbb{A})$ has a fixed spectral dimension: the Archimedean place yields 3 spatial dimensions, and the scale dimension yields 1 time dimension, totaling 4. The other p -adic dimensions are “compactified” as finite trees whose effective dimension is zero at large scales. Other dimensionalities would have either too few or too many p -adic trees to stabilize the vacuum under the RG flow. The 3+1 configuration is the unique stable attractor. [speculative]

8. Experimental Signatures

The framework makes sharp, falsifiable predictions:

1. **Prime echoes in gravitational waves.** Black hole merger echoes arrive at time delays $\Delta t_n = 4M \ln n$ for integers n products of small primes. Amplitudes are proportional to $\mu(n)/n$, where μ is the Möbius function — a distinctive comb of alternating signs. [speculative]
2. **CMB parity violation.** p -adic vacuum fluctuations imprint a small, parity-violating component in the CMB polarization (C_l^{EB} cross-spectrum) with an l -dependence related to prime harmonic sums.
3. **Collider fractal substructure.** At sufficiently high energies, jet substructure may show fractal, prime-number clustering of energy deposits — a signature of the p -adic lattice felt by partons.
4. **Quantum simulation.** The adelic Ising model on a Bruhat-Tits tree can be simulated on a quantum computer using qudits (d -dimensional quantum systems) for each prime, revealing prime-number correlations in the ground state and testing the emergence of the Archimedean continuum.

This would be disconfirmed if: (a) gravitational wave echoes show no prime-number periodicity in a dataset with sufficient signal-to-noise to detect the predicted pattern; (b) CMB C_l^{EB} measurements improve by an order of magnitude and remain consistent with zero across all l ; (c) collider jet substructure at $\gtrsim 100$ TeV remains consistent with QCD shower Monte Carlo predictions without fractal substructure above statistical noise. [speculative]

9. The Moduli Space and the Constants of Nature

The adelic distinction network is not unique; it belongs to a moduli space of possible self-consistent networks, parametrized by the choice of reductive group G and the global adelic moduli. Our universe corresponds to a specific point in this space. The framework suggests a gradient flow on the moduli space toward a universal attractor — a self-organized critical point where distinction entropy is maximized and the product formula is most symmetric.

The values of the dimensionless constants (masses, couplings, mixing angles) are the coordinates of this attractor. They can be computed as local minima of an effective potential on moduli space given by the adelic action. The problem of fine-tuning becomes a problem of dynamical systems on the space of all possible physical laws. The attractor's properties are largely independent of initial conditions, making the observed laws highly probable — a resolution of the anthropic question without invoking a multiverse. [speculative]

This would be disconfirmed if: the effective potential on the moduli space of adelic distinction networks proves to have multiple deep minima rather than a single dominant attractor, implying that the observed constants are not uniquely determined

by the adelic structure.

10. Conclusion

This paper has enumerated the structural consequences of lifting physics from the Archimedean real numbers to the full adèle ring. The idele class group — the “adelic circle” — emerges as the single mathematical object whose automorphic representations generate gauge symmetries, particle species, and cosmological parameters. The product formula $\prod_v |q|_v = 1$ is the invariant that enforces conservation laws, quantizes charge, ensures CPT symmetry, cancels UV divergences, and resolves the cosmological constant problem.

The framework is not a completed theory but a research program. The next steps are computational: derive the Standard Model’s dimensionless parameters as periods of automorphic forms, compute the effective potential on the moduli space of adelic distinction networks, and confront the experimental predictions with data. The adelic perspective does not add new assumptions to physics; it removes one — the unjustified restriction to the real numbers — and allows the mathematics of the adèles to determine what remains.

Declarations

Funding

This work received no external funding.

Competing Interests

The author declares no competing interests.

Data Availability

No experimental data were generated or analyzed in this study.

Author Contributions

Single-author work.

Acknowledgments

This work builds on the mathematical framework established by Ostrowski (1916), Tate (1950), Bruhat-Tits (1972), and the Langlands program, and on prior QNFO papers including the Continuum Trilogy, ODR, and the Non-Anthropocentric Natural Units program.

License

QNFO Unified License Agreement (QNFO-ULA).

Errata

None (first version).

Pre-Registration

The experimental predictions in Section 8 are pre-registered: the specific patterns (prime-number echo delays, Möbius amplitude scaling, CMB C_l^{EB} structure) are stated in advance of the data that would confirm or disconfirm them.

Ethical Statement

This is a theoretical physics paper. No human or animal subjects, no environmental impact, no dual-use concerns.

References

- [1] Ostrowski, A. “Über einige Lösungen der Funktionalgleichung $\varphi(x) \cdot \varphi(y) = \varphi(xy)$.” *Acta Mathematica* 41 (1916): 271–284.
- [2] Spencer Brown, G. *Laws of Form*. George Allen and Unwin, 1969.
- [3] Langlands, R. P. “Automorphic representations, Shimura varieties, and motives.” *Proceedings of Symposia in Pure Mathematics* 33.2 (1979): 205–246.
- [4] Vladimirov, V. S., Volovich, I. V., and Zelenov, E. I. *p-Adic Analysis and Mathematical Physics*. World Scientific, 1994.
- [5] Connes, A. “Trace formula in noncommutative geometry and the zeros of the Riemann zeta function.” *Selecta Mathematica* 5 (1999): 29–106.
- [6] Tate, J. “Fourier analysis in number fields and Hecke’s zeta-functions.” *PhD Thesis*, Princeton University, 1950.
- [7] Dirac, P. A. M. “The quantum theory of the electron.” *Proceedings of the Royal Society A* 117 (1928): 610–624.
- [8] Shannon, C. E. “A mathematical theory of communication.” *Bell System Technical Journal* 27 (1948): 379–423, 623–656.
- [9] Bekenstein, J. D. “Universal upper bound on the entropy-to-energy ratio for bounded systems.” *Physical Review D* 23 (1981): 287–298.

[10] Peccei, R. D. and Quinn, H. R. “CP conservation in the presence of pseudoparticles.” *Physical Review Letters* 38 (1977): 1440–1443.